Class 9

Factories, Firms, and Scale

Part II — Building the American Growth Machine

The American Economy — draft course materials (proposal under discussion)

The point of today's class

America's growth machine did not run on inventions alone — it needed organizations that could coordinate workers, machines, capital, energy, knowledge, logistics, and time at enormous scale.

Why it matters

- Why can McDonald's make the same fries everywhere?
- From artisanal workshop to Ford plant to AI data center, they all share one thing: **systems for coordinating** people, machines, information, timing, quality, and capital.
- Class 8 asked where ideas come from. Today: how firms turn ideas into mass production. Next class: whether workers got paid for it.
- The claim of the hour: **coordination, not just invention, is a source of prosperity**.

Today

1. Why do firms exist?
2. Mechanization: hand vs. machine
3. The managerial corporation
4. The Model T and scale economies
5. Organization as a technology
6. Corporate science and business dynamism

Why do firms exist? (Coase, without the math)

- Why doesn't every worker sell every task separately on the open market?
- Because **using markets is not free**: search, bargaining, contracts, monitoring, enforcement.
- Sometimes hierarchy is cheaper than constant contracting — a restaurant does not auction every task every five minutes.
- Modern echo: Uber, DoorDash, Amazon Marketplace, and gig work are redrawing the boundary between markets and firms.

Artisanal shop	Factory
Skilled craft labor	Division of labor
Human and animal power	Inanimate power
Local markets	National markets
Informal coordination	Rules, schedules, hierarchy

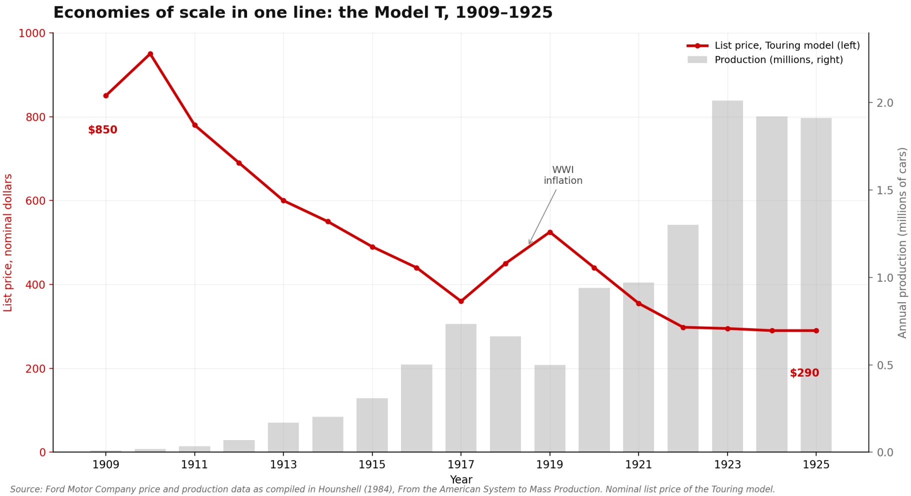
Hand vs. machine labor (Atack, Margo, and Rhode)

- The empirical anchor: the U.S. Department of Labor's mid-1890s *Hand and Machine Labor Study* — a task-level comparison of hand and machine production.
- Shoes, textiles, barrels, nails, printing: mechanized processes were faster, cheaper, more standardized, at far larger scale.
- A machine is not just capital — it is **capital plus energy plus organization** (muscle, water wheel, steam, electric motor, robot arm).
- Mechanization did not simply replace workers: it changed tasks, skills, timing, supervision, and quality — the 1890s version of today's automation debate.

The managerial corporation (Chandler)

- Large enterprises arose with strikingly similar patterns: **organizational capabilities** in production, distribution, purchasing, finance, and research.
- The managerial hierarchy: shareholders, board, executives, divisions, plants, workers.
- **Separation of ownership from control**: it allows scale — and creates agency problems.
- Where scale mattered: railroads (scheduling, finance), steel (fixed costs), oil (refining, pipelines), autos (assembly, dealers), retail (purchasing, inventory), platforms (network effects, data).

Economies of scale in one line: the Model T, 1909–1925



Source: Ford Motor Company price and production data as compiled in Hounshell (1984), From the American System to Mass Production. Nominal list price, Touring model.

What the Model T line shows: scale economies

- The list price falls by roughly two-thirds from 1909 to the mid-1920s — while annual production goes from thousands to millions.
- **Economies of scale**: average cost falls as output rises, because huge fixed costs are spread over more units.
- **Learning-by-doing**: each million cars taught Ford how to build the next million more cheaply.
- The moving assembly line was an invention, a factory system, and a management system *at once*.
- Both sides of the bargain: lower prices, higher output, higher wages in some cases — repetitive work, tighter supervision, labor conflict.

Organization is a technology

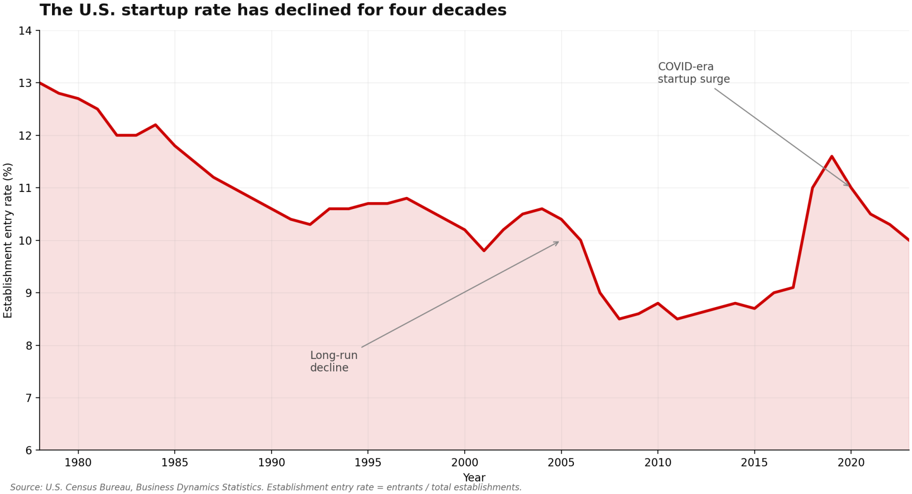
Ordinary technology	Organizational technology
Electric motor	Assembly line
Computer	Inventory system
Truck	Hub-and-spoke logistics
Barcode	Walmart supply chain
AI model	Workflow redesign

- A society can invent a technology and still fail to use it productively if its organizations do not adapt — the GPT lesson from Class 8.
- Henderson–Clark, **architectural innovation**: a firm can know every component and still fail when how the components fit together changes (film to digital, stores to e-commerce).

Big organizations: rules, labs, and limits

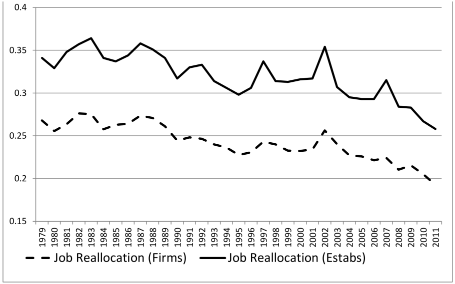
- Rules are a coordination technology too: checklists, standards, and protocols encode safety, learning, and accountability.
- But rules can become obsolete, captured, or excessive — the question is never "rules good or bad" but *what problem does this rule solve, and is it still worth the cost?*
- Bell Labs: corporate science at scale — the transistor, information theory, lasers, Unix.
- Arora, Belenzon, and Patacconi: large firms have retreated from science since 1980 — publications by company scientists fell while patenting stayed stable.
- Not all productive organizations are private firms: DARPA, NASA, Operation Warp Speed, hospitals, universities. The question is what organizational form solves the coordination problem.

The declining U.S. startup rate



Source: U.S. Census Bureau, Business Dynamics Statistics.

Job reallocation and the productivity dividend



Notes to Figure 3:

1. Both series pertain to the nonfarm private sector of the U.S. economy. The job reallocation rate across establishments is the sum of March-to-March absolute employment changes summed over entering, expanding, exiting and shrinking establishments, expressed as a fraction of employment. The job reallocation rate across firms is defined analogously based on firm-level employment changes.
2. The plotted series are from the Business Dynamic Statistics program and tabulations on the Longitudinal Business Database by Decker et al. (2014b).

Source: Decker, Haltiwanger, Jarmin, and Miranda (2014), Journal of Economic Perspectives.

Discussion

Four production systems: a local bakery, a Ford assembly plant, the Walmart distribution system, and an AI model company. For each: what are the fixed costs? What is standardized? What must be coordinated? What does the firm do better than the market? And who gains — consumers, workers, owners, or all three?

Read before next class

- Atack, Margo, and Rhode (2019), "'Automation' of Manufacturing in the Late Nineteenth Century: The Hand and Machine Labor Study," *JEP* (skim).
- Chandler (1992), "Organizational Capabilities and the Economic History of the Industrial Enterprise," *JEP* (skim).